Mini Intrusion Detection System (IDS)

1. Introduction

A Mini Intrusion Detection System (IDS) is a basic software tool developed to monitor system log files and automatically detect suspicious or malicious activities. It acts as a security mechanism that helps identify potential threats by analysing recorded system events.

In any computer system, log files are generated continuously to store information about user activities, login attempts, commands executed, and system operations. These logs provide valuable insights into how the system is being used and can help in identifying abnormal or unauthorized behaviour.

Instead of manually checking large volumes of log data, this project provides an automated solution that processes log entries and detects patterns that may indicate cyber-attacks. The system reads logs from a file named logs.txt, processes them one by one, and analyses each entry for suspicious content.

The Mini IDS focuses on detecting common types of threats such as repeated failed login attempts, SQL injection patterns, and execution of dangerous system commands. Whenever such activities are found, the system generates alerts to notify the user about potential security risks.

This project is useful for understanding how intrusion detection systems work at a basic level and demonstrates how programming and data structures can be applied to solve real-world cybersecurity problems.

2. Aim of the Project

The main aim of this project is to design and implement a system that can automatically analyse system logs and detect suspicious or malicious activities effectively.

The system focuses on detecting three major types of threats:

- Brute force login attacks
- SQL injection attacks
- Dangerous system command execution

The project is designed to be simple, understandable, and practical for learning purposes.

3. Problem Statement

System logs are generated continuously but are often not actively monitored. This creates a security risk because attackers can perform malicious activities without being detected.

Manual log analysis is time-consuming and inefficient, especially when dealing with large volumes of data.

There is a need for an automated system that can analyze logs quickly and detect suspicious activities accurately.

This project aims to solve this problem by developing a Mini IDS that processes logs and generates alerts automatically.

4. Objectives

- To read and process log files automatically
- To detect brute force login attempts
- To identify SQL injection patterns
- To detect dangerous system commands
- To generate alerts for suspicious activities
- To use data structures efficiently
- To understand practical cybersecurity implementation

5. Scope of the Project

- The system works on log files stored in the system
- It detects predefined attack patterns
- It si suitable for academic and small-scale use
- It demonstrates basic IDS functionality
- It can be extended for real-time monitoring in future

6. What the System Detects

6.1 Brute Force Login Attack

- Multiple failed login attempts from the same IP address
- Count of failed attempts is tracked
- If attempts ≥ 3 → attack detected
- Helps prevent unauthorized access

6.2 SQL Injection Attack

- Malicious SQL queries used to manipulate databases
- Examples:
    - ' OR '1'='1
    - DROP TABLE
    - UNION SELECT
- Detected using pattern matching
- Generates alert for database attacks

6.3 Dangerous Command Execution

- Harmful system commands executed by attackers
- Examples:
    - sudo rm -rf
    - shutdown
    - wget malicious
- Detected as suspicious patterns
- Indicates serious system threat

7. Data Structures Used

7.1 Queue

- Stores log entries
- Processes logs in FIFO order
- Maintains sequence of events

7.2 Hash Map (unordered_map)

- Stores IP → failed login count
- Provides fast lookup
- Detects brute force attacks efficiently

7.3 Trie (Prefix Tree)

- Stores suspicious patterns
- Enables fast pattern searching
- Efficient for multiple pattern detection

8. System Architecture / Flow

- Read logs from logs.txt
- Store logs in Queue
- Process logs sequentially
- Extract IP address
- Check failed login attempts
- Update Hash Map
- If count ≥ 3 → brute force alert
- Check patterns using Trie
- If pattern found → suspicious alert
- Continue until all logs processed

9. Working of the System

- Logs are read from file and stored in Queue
- Each log is processed one by one
- IP address is extracted
- Failed login attempts are counted
- If threshold reached → brute force alert
- Log is checked for suspicious patterns
- If pattern found → alert generated
- Process continues until all logs are analyzed

10. Explanation of the Code

Header Files

- Used for input-output, file handling, and data structures

TrieNode Class

- Represents node of Trie
- Stores characters and end-of-word flag

Trie Class

- Handles insertion and searching of patterns

extractIP Function

- Extracts IP address from log

Main Function

- Controls execution
- Reads logs
- Processes logs
- Detects attacks
- Displays alerts

11. Sample Input

- Log file contains timestamp, IP address, user, and activity
- Includes login attempts, commands, and queries

12. Expected Output

- The system generates alerts based on detected suspicious activities such as brute force attacks and malicious commands.
- The following figure shows the output of the Mini IDS after processing the log file.

```
----- MINI IDS STARTED -----

Processing Log: 2026-04-30 12:00:00 IP=192.168.1.10 User=admin Failed login

Processing Log: 2026-04-30 12:00:02 IP=192.168.1.10 User=admin Failed login

Processing Log: 2026-04-30 12:00:05 IP=192.168.1.10 User=admin Failed login
[ALERT] Brute Force Attack Detected from IP: 192.168.1.10

Processing Log: 2026-04-30 12:01:17 IP=10.0.0.5 User=guest Command=sudo rm -rf /
[ALERT] Suspicious Activity Detected from IP: 10.0.0.5

Processing Log: 2026-04-30 12:02:30 IP=10.0.0.6 User=web Query=' OR '1'='1
[ALERT] Suspicious Activity Detected from IP: 10.0.0.6

----- IDS SCAN COMPLETED -----
PS C:\Users\prasa\Downloads\Coding Skills Project\Mini_IDS>
```

Figure 1

Figure 1: Output of Mini Intrusion Detection System

- The output shows how the system processes each log entry step by step.
- After detecting three failed login attempts from the same IP address, it generates a brute force attack alert.
- It also detects dangerous commands and SQL injection patterns and produces corresponding alerts.
- Finally, the system displays a completion message indicating that all logs have been processed successfully.

13. Time and Space Complexity

- Log reading → O(n)
- Log processing → O(n)
- Hash Map operations → O(1) average
- Trie insertion → O(m)
- Pattern searching → efficient compared to normal string search

Where:

- n = number of logs
- m = pattern length

14. Advantages

- Simple and easy to understand
- Fast detection using efficient data structures
- Maintains correct processing order
- Easy to extend with new attack patterns
- Demonstrates real-world application of DSA

15. Limitations

- Detects only predefined patterns
- Cannot detect unknown attacks
- Works on static logs only
- Limited scalability

16. Future Improvements

- Real-time log monitoring
- Machine learning-based detection
- Graphical user interface
- Alert storage and reporting
- Email/SMS notification system
- Integration with network security tools

17. Conclusion

The Mini Intrusion Detection System is a simple yet effective project that demonstrates how system logs can be analyzed to detect suspicious activities automatically.

It successfully detects brute force login attacks, SQL injection attempts, and dangerous command execution using Queue, Hash Map, and Trie data structures.

The project provides a strong foundation for understanding cybersecurity concepts and practical implementation of data structures in real-world applications.